

# LOCALIZED BOST–CONNES SYSTEMS IV: NONCOMMUTATIVE METRIC GEOMETRY

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ABSTRACT. Three invariants of the localized Bost–Connes system jump by a fixed amount at each phase transition: the rank of the obstruction group jumps by one [9], the subfactor index doubles, and the relative entropy of an extremal phase against the symmetric one increases by  $\log 2$  (Papers II and III of the series). We show that the natural *metric* invariant does not: for a horizontal Dirac operator on the base algebra  $C(X)$  of the system, the Connes–Rieffel distance between the measures of the two symmetry-broken extremal  $\text{KMS}_\beta$  states vanishes *continuously* as  $\beta$  decreases to the threshold, so the algebraic transition is of first order and the metric transition of second.

The reason is a rigidity. For a horizontal Dirac operator,  $C(X)$  is a compact quantum metric space in Rieffel’s sense if and only if the metric weights are summable, and summability forces the order parameter  $\Psi$  out of the Lipschitz domain: the finite truncations  $\Psi_N^{(m)}$  have commutator norms diverging like  $(\sum_{\ell \leq N} w_{\ell,m}^{-2})^{1/2}$ . A macroscopic tail observable can never be smoothed into the Lipschitz algebra of a Dirac operator that metrizes the weak-\* topology, and it is exactly this failure that makes the metric transition continuous: the supremum defining the distance is carried by finite-level functions whose overlap with  $\Psi$  degenerates at the threshold.

Quantitatively we prove the squeeze

$$w_0 \Pi_\infty(\beta) \leq W_1(\varphi_+, \varphi_-) \leq w_0, \quad \Pi_\infty(\beta) = \prod_{\ell \in R} \frac{1 - \ell^{-\beta}}{1 + \ell^{-\beta}},$$

with  $w_0$  the weight of the single elementary move at the component 3. The lower bound is the Kakutani affinity of the parity family, positive precisely when  $\sum_{\ell \in R} \ell^{-\beta} < \infty$ , so the Borel–Cantelli criterion governing the existence of  $\Psi$  reappears verbatim as a metric quantity. The distance is therefore *local*, bounded by a single-coordinate cost, although  $\Psi$  is a tail observable and  $\nu_+ \perp \nu_-$  at maximal total variation. One structural fact is recorded along the way: the modular direction is metrically blind, any Dirac operator commuting with  $C(Y_S)$  giving infinite distance, so that a metric on the  $\text{KMS}_\beta$  simplex must come from the horizontal geometry of the base; whether our horizontal operator extends to a spectral triple on the crossed product is left open.

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Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/IV-metric/](#).

## 1. INTRODUCTION: THREE JUMPS AND ONE SLOPE

We use the notation of [8]. In the symmetry-broken phase of [8, §4.2] —  $K = \mathbb{Q}$ ,  $S = \{3\} \cup P_1 \cup R$  with  $P_1 = \{\ell \equiv 1 \pmod{3}\}$  and  $R$  infinite of primes  $\equiv 2 \pmod{3}$  — the  $\text{KMS}_\beta$  simplex is a segment as soon as  $\sum_{\ell \in R} \ell^{-\beta} < \infty$ , its extreme points  $\varphi_\pm$  separated by

$$(1.1) \quad \Psi(x) = \left( \frac{u_3(x) \bmod 3}{3} \right) (-1)^{\sum_{\ell \in R} v_\ell(x)} \in \{\pm 1\}, \quad \nu_\pm = 2 \mathbf{1}_{\{\Psi=\pm 1\}} \nu_{\text{sym}},$$

so that  $\nu_+ - \nu_- = 2\Psi \nu_{\text{sym}}$  [8, Prop. 4.12]; here  $\Psi$  is defined  $\nu_{\text{sym}}$ -almost everywhere exactly when  $\sum_{\ell \in R} \ell^{-\beta} < \infty$ .

*The series.* The present paper is IV. The foundational paper is [8]; [9] is I and treats the factor type and the spectrum of transitions. Papers II (ideals, boundary and index), III ( $\Xi_\beta$  as an invariant of a measure class) and V (the free variant) are in preparation; nothing here depends on them, and we refer to them by numeral, without citation.

At a threshold  $\beta_n$  the companion papers record three discontinuities: the rank of  $\Xi_\beta$  increases by one and the normalized order parameter is a pure jump function [9, Thm. 6.5]; the index  $[M_{\text{sym}} : M_\varepsilon]$  of the symmetry-breaking inclusion doubles (Paper II); and the relative entropy  $S(\omega_{\varphi_\varepsilon} \| \omega_{\varphi_{\text{sym}}})$  increases by  $\log 2$  (Paper III). All three count phases, and phases are counted in integers.

The metric invariant behaves differently, and Theorem 6.2 is the reason for this paper. The mechanism is Theorem 4.2:  $\Psi$  cannot be Lipschitz for a Dirac operator that metrizes the weak-\* topology, so the supremum defining the distance cannot be attained on  $\Psi$  itself and is carried instead by finite-level approximations whose correlation with  $\Psi$  degenerates at the threshold.

## 2. THE BASE SPACE AND THE HORIZONTAL DIRAC OPERATOR

## 2.1. Two structural facts.

**Proposition 2.1** (The modular direction is metrically blind). *Let  $\mathcal{A}_{K,S}$  be represented on a Hilbert space and  $D$  a self-adjoint operator commuting with the image of  $C(Y_S)$  — for instance, in a representation in which  $C(Y_S)$  acts diagonally with respect to a basis  $(\epsilon_a)_{a \in J_S}$ , the modular operator  $D_{\text{mod}} \epsilon_a = (\log N_a) \epsilon_a$ , which generates the dynamics. Then the Connes distance [1, 2] satisfies  $W_1^D(\varphi_+, \varphi_-) = +\infty$  whenever  $\varphi_+ \neq \varphi_-$ .*

*Proof.* Both states are of the form  $\nu_\pm \circ E$  with  $E$  the canonical expectation, so they agree on every  $\mu_a f \mu_b^*$  with  $a \neq b$ ; hence  $\varphi_+ \neq \varphi_-$  forces  $\varphi_+(f) \neq \varphi_-(f)$  for some  $f \in C(Y_S)$ . For such  $f$  and every  $s > 0$  we have  $[D, sf] = 0$ , so  $sf$  lies in the Lipschitz ball and  $|\varphi_+(sf) - \varphi_-(sf)| \rightarrow \infty$ .  $\square$

*Remark 2.2* (Why we work on the base). Proposition 2.1 says that a metric on the  $\text{KMS}_\beta$  simplex must come from an operator that does *not* commute with  $C(Y_S)$ , i.e. from the horizontal geometry of the base. Every  $\text{KMS}_\beta$  state is  $\nu \circ E$  for a measure  $\nu$  on  $Y_S$  and the canonical expectation  $E$  [8, Prop. 2.8], and  $\varphi \mapsto \nu$  is an affine bijection onto the scaling-invariant measures. We therefore put a Connes–Rieffel metric on the base algebra and measure the simplex through it:

$$W_1(\varphi_1, \varphi_2) := W_1^{C(X)}(\nu_1, \nu_2) = \sup \{ |\nu_1(f) - \nu_2(f)| : f \in C(X)_{sa}, L(f) \leq 1 \},$$

with  $L$  the seminorm of Lemma 2.5. If a Dirac operator  $D$  on a representation of  $\mathcal{A}_{K,S}$  restricted to  $C(X)$  gave the seminorm  $L$  and had bounded commutators with the isometries, one would have  $W_1^{\mathcal{A}_{K,S}} \geq W_1^{C(X)}$  for the states  $\nu_i \circ E$ , with equality when the gauge action is implemented by unitaries commuting with  $D$ . The operator of Definition 2.4 does not have this property: it must weight the level  $k$  of each coordinate  $v_\ell$  by a summable sequence  $w_{\ell,k}$  (Theorem 3.2), whereas the isometry  $\mu_p$  shifts the levels and would force  $w_{\ell,k} = w_{\ell,k+1}$ . We make no claim about a spectral triple on the crossed product; see the assessment.

**2.2. The compact model and the moves.**  $Y_S$  is a quotient of  $\widehat{\mathcal{O}}_S$  in which the coordinates  $(v_\ell, u_\ell)$  are a measure-theoretic rather than topological decomposition. We therefore work on

$$(2.1) \quad X := G_S \times \overline{\mathbb{N}}^S, \quad \overline{\mathbb{N}} = \mathbb{N} \cup \{\infty\},$$

compact metrizable and totally disconnected, on which by [8, Lem. 2.10]

$$(2.2) \quad \nu_{\text{sym}} = m_{G_S} \otimes \bigotimes_{\ell \in S} \mu_{t_\ell}, \quad \mu_t(k) = (1-t)t^k, \quad t_\ell = \ell^{-\beta},$$

so  $\nu_{\text{sym}}$  charges no hyperplane  $\{v_\ell = \infty\}$  and the natural map  $X \rightarrow Y_S$  is a measure isomorphism off a null set. All test functions constructed below lie in  $C(X)$ .

**Definition 2.3** (Moves and weights). Let  $g_0 \in G_S$  be the unit whose 3-component is  $-1$  and whose other components are 1, and  $\theta_0 = \theta_{g_0}$  the corresponding involution of  $X$  (translation of the  $G_S$ -coordinate by  $g_0$ ); it flips  $\chi_3$ . Let  $G_S = U_0 \supset U_1 \supset U_2 \supset \dots$  be open subgroups with  $U_1 = \ker \chi_3$  and  $\bigcap_k U_k = \{1\}$ , and let  $(g_i)_{i \geq 1}$  be elements of  $U_1$  such that, for every  $k \geq 1$ , the  $g_i$  lying in  $U_k$  generate  $U_k/U_{k+1}$ ; write  $\theta_i = \theta_{g_i}$ , so that  $\{g_0\} \cup \{g_i\}$  generates a dense subgroup of  $G_S$  and every element of  $U_k$  is a convergent product of  $g_i^{\pm 1}$  with  $g_i \in U_k$ . For  $\ell \in S$ ,  $k \in \mathbb{N}$  let  $\tau_{\ell,k}$  be the *localized shift* sending  $(u, v) \mapsto (u, v + e_\ell)$  if  $v_\ell = k$  and fixing everything otherwise. Index these moves by  $j \in J$ , write  $\Delta_j f = f \circ \vartheta_j - f$ , and fix weights  $w_j > 0$  with

$$(2.3) \quad \Sigma := \sum_{j \in J} w_j < \infty,$$

$w_{\ell,k}$  non-increasing in  $k$ . The weights are *independent of*  $\beta$ ; only the measure depends on  $\beta$ . Write  $w_0$  for the weight of  $\theta_0$ .

**Definition 2.4.** On  $\mathcal{H} = L^2(X, \nu_{\text{sym}}) \oplus (L^2(X, \nu_{\text{sym}}) \otimes \ell^2(J))$  put  $D_{\text{base}} = \begin{pmatrix} 0 & \delta^* \\ \delta & 0 \end{pmatrix}$  with  $(\delta\xi)_j = w_j^{-1} \Delta_j \xi$ , and let  $C(X)$  act diagonally by multiplication.

**Lemma 2.5** (Commutator formula). *For  $f \in C(X)$ ,*

$$(2.4) \quad L(f) := \|[D_{\text{base}}, f]\| = \left\| \left( \sum_{j \in J} w_j^{-2} |\Delta_j f|^2 \right)^{1/2} \right\|_\infty,$$

a seminorm with  $L(f^*) = L(f)$  and  $L(fg) \leq \|f\|_\infty L(g) + \|g\|_\infty L(f)$ . The operator  $D_{\text{base}}$  is of the type introduced by Christensen and Ivan for AF algebras and Cantor sets [3], with the moves in the role of the edges.

*Proof.*  $(\delta M_f \xi)_j - (M_f^\oplus \delta \xi)_j = w_j^{-1} (\Delta_j f)(\xi \circ \vartheta_j)$ , so  $[D_{\text{base}}, M_f]$  is off-diagonal with multiplier of pointwise  $\ell^2(J)$ -norm as displayed. The algebraic properties follow from  $\Delta_j(fg) = (f \circ \vartheta_j) \Delta_j g + (\Delta_j f)g$ .  $\square$

### 3. $C(X)$ IS A COMPACT QUANTUM METRIC SPACE

**Lemma 3.1** (Kernel and paths).  $\{f \in C(X) : L(f) = 0\} = \mathbb{C}1$ ; and for  $L(f) \leq 1$ ,

$$|f(x) - f(y)| \leq \sum_{j \in J(x,y)} w_j$$

for any finite or convergent sequence of moves carrying  $x$  to  $y$  using each at most once. In particular  $\text{diam} \leq \Sigma < \infty$ . Let  $J_{\leq k}$  consist of  $\theta_0$ , the  $\theta_i$  with  $g_i \notin U_k$ , and the  $\tau_{\ell,m}$  with  $\ell$  among the first  $k$  primes of  $S$  and  $m < k$ . If  $x = (u, v)$  and  $y = (u', v')$  satisfy  $u'u^{-1} \in U_k$  and  $v_\ell = v'_\ell$  for the first  $k$  primes  $\ell$  whenever  $\min(v_\ell, v'_\ell) < k$ , then

$$|f(x) - f(y)| \leq \Sigma_k := \sum_{j \notin J_{\leq k}} w_j \xrightarrow[k \rightarrow \infty]{} 0.$$

*Proof.*  $L(f) = 0$  forces  $\Delta_j f \equiv 0$ ; invariance under all  $\theta_i$  gives, by density and continuity, independence of  $u$ , and invariance under all  $\tau_{\ell,k}$  gives constancy on the finitely supported  $v$ , dense in  $\overline{\mathbb{N}}^S$ . For the estimate,  $L(f) \leq 1$  gives  $|f(\vartheta_j x) - f(x)| \leq w_j$  for each single move; along a finite sequence of distinct moves the differences add, and a convergent sequence is handled by continuity of  $f$ . The localized shifts make “each move at most once” achievable: passing from  $v_\ell = m$  to  $v_\ell = m'$  uses the distinct moves  $\tau_{\ell,m}, \dots, \tau_{\ell,m'-1}$ , and  $v_\ell = \infty$  is reached by the convergent sequence  $\tau_{\ell,m}, \tau_{\ell,m+1}, \dots$ . For the last statement, carry  $x$  to  $y$  by first writing  $u'u^{-1}$  as a convergent product of  $g_i^{\pm 1}$  with  $g_i \in U_k$  (possible by the choice of the  $g_i$ ), then adjusting the  $v_\ell$  with  $\ell$  beyond the first  $k$  primes, and finally the  $v_\ell$  of the first  $k$  primes, which by hypothesis differ only at levels  $\geq k$ : every move used lies outside  $J_{\leq k}$ , and the moves used are distinct.  $\square$

**Theorem 3.2** (Compact quantum metric space). *Under (2.3),  $(C(X), L)$  is a compact quantum metric space in the sense of Rieffel [6, 7]:  $L$  is a lower semicontinuous Lip-norm with kernel  $\mathbb{C}1$ , finite on the dense subalgebra of functions depending on finitely many coordinates and eventually constant in each  $v_\ell$ , and the Monge–Kantorovich metric*

$$W_1(\varphi_1, \varphi_2) = \sup\{|\varphi_1(f) - \varphi_2(f)| : f \in C(X)_{sa}, L(f) \leq 1\}$$

*metrizes the weak-\* topology on the state space, of diameter  $\leq \Sigma$ .*

*Proof.* By Rieffel’s criterion [6, Thm. 1.8] it suffices that  $\mathcal{B} = \{f \in C(X)_{sa} : L(f) \leq 1, f(x_0) = 0\}$  be totally bounded in  $\|\cdot\|_\infty$ . Lemma 3.1 gives  $\|f\|_\infty \leq \Sigma$  and equicontinuity: the sets  $\{y : u'u^{-1} \in U_k, v'_\ell = v_\ell \text{ or } \min(v_\ell, v'_\ell) \geq k \text{ for the first } k \text{ primes}\}$  are neighbourhoods of  $x$  on which every  $f \in \mathcal{B}$  varies by at most  $\Sigma_k \rightarrow 0$ ;  $X$  is compact, so Arzelà–Ascoli applies. Lower semicontinuity follows from (2.4), each  $f \mapsto \Delta_j f$  being continuous for uniform convergence.  $\square$

*Remark 3.3* (Summability is necessary). Suppose  $\sum_k w_{\ell,k} = \infty$  for some  $\ell$ , and put  $f_n = \sum_{k < n} w_{\ell,k} \mathbf{1}_{\{v_\ell > k\}}$ . Then  $\Delta_{\ell,k} f_n = w_{\ell,k}$  for  $k < n$ , no other elementary difference is nonzero,  $L(f_n) = 1$ , and  $\text{osc}(f_n) = \sum_{k < n} w_{\ell,k} \rightarrow \infty$ , so  $\mathcal{B}$  is unbounded; the same construction along a level  $U_k/U_{k+1}$  of the  $G_S$ -moves shows that  $\sum_i w_i = \infty$  is likewise fatal. Summability of the weights and finiteness of the diameter are equivalent here.

#### 4. THE ORDER PARAMETER IS NOT LIPSCHITZ

**Lemma 4.1** (Elementary differences). *On the conull set where  $\Psi$  is defined,  $\Delta_0 \Psi = -2\Psi$ ,  $\Delta_{\ell,k} \Psi = -2\Psi \cdot \mathbf{1}_{\{v_\ell = k\}}$  for  $\ell \in R$ , and  $\Delta_j \Psi = 0$  for all other moves.*

*Proof.*  $\theta_0$  multiplies  $u_3$  by a non-residue, flipping  $\chi_3$  and hence  $\Psi$ ;  $\tau_{\ell,k}$  increases  $v_\ell$  by one exactly on  $\{v_\ell = k\}$ , flipping the parity factor when  $\ell \in R$ . The  $\theta_i, i \geq 1$ , lie in  $\ker \chi_3$ .  $\square$

**Theorem 4.2** (Exclusivity). *Assume (2.3) and  $R \neq \emptyset$ . Then  $L(\Psi) = \infty$ , and indeed  $L(\Psi_N) = \infty$  for every  $N \geq \min R$ , where  $\Psi_N = \chi_3 \prod_{\ell \in R, \ell \leq N} (-1)^{v_\ell}$ ,  $L$  being extended to bounded Borel functions by the same formula (2.4). No truncation of the order parameter beyond the level of  $\chi_3$  lies in the Lipschitz domain of the operators of Definition 2.4; and since  $\Psi_N$  is discontinuous along  $\{v_\ell = \infty\}$  for  $\ell \in R, \ell \leq N$ , it lies in the Lipschitz domain of no Lip-norm on  $C(X)$  whatever.*

*Proof.* By Lemma 4.1, at a point  $x$  with  $v_\ell(x) = k$ ,

$$\sum_j w_j^{-2} |\Delta_j \Psi(x)|^2 = 4w_0^{-2} + 4 \sum_{\ell \in R} w_{\ell, v_\ell(x)}^{-2} \geq 4w_{\ell,k}^{-2}.$$

Summability forces  $w_{\ell,k} \rightarrow 0$  as  $k \rightarrow \infty$  for each  $\ell$ , while  $\nu_{\text{sym}}(\{v_\ell = k\}) = (1 - t_\ell)t_\ell^k > 0$  for every  $k$ ; so the essential supremum in (2.4) is  $+\infty$ . The same computation truncated at  $N$  gives the second assertion.  $\square$

**Corollary 4.3** (Quantitative divergence). *Let  $\phi_m(v) = (-1)^v$  for  $v \leq m$  and  $(-1)^m$  for  $v > m$ , so  $\phi_m \in C(\overline{\mathbb{N}})$ , and  $\Psi_N^{(m)} = \chi_3 \prod_{\ell \in R, \ell \leq N} \phi_m(v_\ell) \in C(X)$ . Then*

$$(4.1) \quad L(\Psi_N^{(m)}) = 2 \left( w_0^{-2} + \sum_{\ell \in R, \ell \leq N} w_{\ell, m-1}^{-2} \right)^{1/2} \xrightarrow{m \rightarrow \infty} \infty,$$

while  $\|\Psi_N^{(m)} - \Psi_N\|_2^2 \leq 4 \sum_{\ell \in R, \ell \leq N} t_\ell^m$ .

*Proof.*  $\Delta_{\ell, k} \phi_m = 0$  for  $k \geq m$  and  $= \mp 2$  for  $k < m$ ; since  $w_{\ell, k}$  is non-increasing the supremum over  $k < m$  of  $w_{\ell, k}^{-2}$  is  $w_{\ell, m-1}^{-2}$ , attained on a set of positive measure. The  $L^2$  estimate is the union bound on  $\{v_\ell > m\}$ .  $\square$

*Remark 4.4* (Interpretation). The macroscopic charge has infinite impedance at the microscopic scale. Resolving it to depth  $m$  costs a Lipschitz seminorm growing like  $w_{\ell, m-1}^{-1}$ , and full resolution costs infinitely much — because a metric that is to induce the weak-\* topology must assign summable, hence vanishing, weights to deep coordinates. A tail observable is by construction insensitive to any finite set of coordinates; a compact quantum metric must be insensitive, in the opposite sense, to all but finitely many. The two requirements are incompatible.

## 5. THE SQUEEZE THEOREM

Throughout this section  $\sum_{\ell \in R} \ell^{-\beta} < \infty$ , so  $\Psi$  exists and  $\varphi_+ \neq \varphi_-$ . Put

$$(5.1) \quad m_\ell = \frac{1 - t_\ell}{1 + t_\ell} = \mathbb{E}_{\mu_{t_\ell}}[(-1)^{v_\ell}], \quad \Pi_\infty = \prod_{\ell \in R} m_\ell.$$

**Lemma 5.1** (Kakutani criterion).  $0 < m_\ell < 1$ , and  $\Pi_\infty > 0 \iff \sum_{\ell \in R} (1 - m_\ell) < \infty \iff \sum_{\ell \in R} \ell^{-\beta} < \infty$ .

*Proof.*  $1 - m_\ell = 2t_\ell/(1 + t_\ell) \in [t_\ell, 2t_\ell]$ , so the two series converge together; an infinite product of factors in  $(0, 1)$  is positive iff the defects are summable.  $\square$

**Lemma 5.2** (Reduction to odd functions).  $L$  is invariant under  $f \mapsto f \circ \theta_0$ ; the odd part  $f^- = \frac{1}{2}(f - f \circ \theta_0)$  satisfies  $L(f^-) \leq L(f)$  and  $\langle f^-, \Psi \rangle = \langle f, \Psi \rangle$ . Every odd  $f$  is  $\chi_3 h$  with  $h = \chi_3 f$  even, and then  $\Delta_0 f = -2f$  and  $\langle f, \Psi \rangle = \langle h, \sigma \rangle$  with  $\sigma = \prod_{\ell \in R} (-1)^{v_\ell}$ .

*Proof.*  $\theta_0$  is a measure-preserving involution commuting with every  $\vartheta_j$ , so  $L$  is  $\theta_0$ -invariant and  $L(f^-) \leq L(f)$ ; and  $\Psi \circ \theta_0 = -\Psi$  gives  $\langle f \circ \theta_0, \Psi \rangle = -\langle f, \Psi \rangle$ . For odd  $f$ ,  $h = \chi_3 f$  is even, and  $\Psi = \chi_3 \sigma$  with  $\chi_3^2 = 1$ .  $\square$

**Theorem 5.3** (Squeeze).

$$(5.2) \quad w_0 \Pi_\infty(\beta) \leq W_1(\varphi_+, \varphi_-) \leq w_0.$$

In particular  $W_1(\varphi_+, \varphi_-) > 0$  if and only if  $\sum_{\ell \in R} \ell^{-\beta} < \infty$ .

*Proof.* By Remark 2.2 and (1.1),  $W_1(\varphi_+, \varphi_-) = \sup\{|\nu_+(f) - \nu_-(f)| : f \in C(X)_{sa}, L(f) \leq 1\} = \sup\{2\langle f, \Psi \rangle : f \in C(X)_{sa}, L(f) \leq 1\}$ , where  $\langle f, \Psi \rangle = \int f \Psi d\nu_{\text{sym}}$ .

*Lower bound.* Take  $f = \frac{w_0}{2} \chi_3$ , continuous and depending only on  $u_3$ . By Lemma 4.1 its only nonzero elementary difference is  $\Delta_0 f = -w_0 \chi_3$ , so  $L(f) = w_0^{-1} w_0 = 1$ ; and  $\chi_3$  and  $\sigma$  being independent under (2.2),

$$2\langle f, \Psi \rangle = w_0 \langle \chi_3, \chi_3 \sigma \rangle = w_0 \mathbb{E}[\sigma] = w_0 \Pi_\infty.$$

*Upper bound.* By Lemma 5.2 take  $f = \chi_3 h$  odd with  $L(f) \leq 1$ . Then  $|\Delta_0 f| = 2|h|$  pointwise, so (2.4) gives  $2w_0^{-1} \|h\|_\infty \leq 1$ ; since  $|\sigma| = 1$ ,  $2\langle f, \Psi \rangle = 2\langle h, \sigma \rangle \leq 2\|h\|_\infty \leq w_0$ .

If instead  $\sum_{\ell \in R} \ell^{-\beta} = \infty$  then  $\Psi$  is undefined a.e.,  $\nu_+ = \nu_-$  and  $W_1 = 0$ ; otherwise  $\Pi_\infty > 0$  by Lemma 5.1.  $\square$

*Remark 5.4* (Locality versus tail behaviour). The upper bound is the cost of the *single* elementary move  $\theta_0$  at the component 3, and its source is that  $\varphi_- = \varphi_+ \circ \theta_0$ : the two phases lie on one  $G_S$ -orbit and the group element realizing the passage is supported at one coordinate. The metric separation is therefore local, even though the separating observable is a tail function and the measures are mutually singular, at maximal total variation distance 2. Total variation sees the tail; the horizontal distance sees the shortest move.

## 6. THE METRIC TRANSITION IS CONTINUOUS

**Lemma 6.1** (Poincaré inequality for the geometric law). *For  $0 < t < 1$ ,  $\mu_t(k) = (1-t)t^k$  and  $\Delta h(k) = h(k+1) - h(k)$ ,*

$$(6.1) \quad \text{Var}_{\mu_t}(h) \leq C(t) \mathbb{E}_{\mu_t} |\Delta h|^2, \quad C(t) = \frac{4t}{(1-t)^2},$$

and the constant is sharp up to the factor 4:  $h(k) = k$  gives equality with  $t/(1-t)^2$ .

*Proof.* This is the discrete weighted Hardy inequality of Muckenhoupt [5] in the form given by Miclo [4] for a birth–death chain with conductances  $c_k = \mu_t(k)$ , applied to  $h - h(0)$ : the optimal constant lies between  $B$  and  $4B$  with

$$B = \sup_{n \geq 1} \left( \sum_{k \geq n} \mu_t(k) \right) \left( \sum_{k=0}^{n-1} c_k^{-1} \right) = \sup_{n \geq 1} \frac{t^n}{1-t} \cdot \frac{t^{-n} - 1}{t^{-1} - 1} = \sup_{n \geq 1} \frac{1 - t^n}{(1-t)(t^{-1} - 1)} = \frac{t}{(1-t)^2}.$$

For  $h(k) = k$ ,  $\text{Var} = t/(1-t)^2$  and  $\mathbb{E}|\Delta h|^2 = 1$ .  $\square$

**Theorem 6.2** (Continuous vanishing at  $\beta_c$ ). *Assume the weights satisfy (2.3) and are bounded, let  $\beta_c = \beta_c(R) > 0$  be the convergence exponent of  $\sum_{\ell \in R} \ell^{-\beta}$ , and assume the series diverges at the exponent itself,  $\sum_{\ell \in R} \ell^{-\beta_c} = \infty$ , as it does for the cascade sets of [8, Prop. 4.16] and [9, Lem. 5.4]. Then*

$$\lim_{\beta \downarrow \beta_c} W_1(\varphi_+, \varphi_-) = 0.$$

Consequently  $\beta \mapsto \text{diam}_{W_1}(\text{KMS}_\beta)$  vanishes on  $(0, \beta_c]$ , is strictly positive on  $(\beta_c, \infty)$ , and is continuous at  $\beta_c$ . (If instead  $\sum_{\ell \in R} \ell^{-\beta_c} < \infty$ , then  $\Psi$  exists at  $\beta_c$  and Theorem 5.3 gives  $W_1 \geq w_0 \Pi_\infty(\beta_c) > 0$  there, while  $W_1 = 0$  for  $\beta < \beta_c$ : the diameter then jumps at  $\beta_c$ , exactly as the algebraic invariants do.)

*Proof.* Let  $f = \chi_3 h$  be odd with  $L(f) \leq 1$ , so  $\|h\|_\infty \leq w_0/2$  by the proof of Theorem 5.3. Let  $\mathcal{F}_N = \sigma(u; v_\ell, \ell \leq N)$  and  $E_N$  the conditional expectation, and split  $2\langle h, \sigma \rangle = 2\langle E_N h, \sigma \rangle + 2\langle h - E_N h, \sigma \rangle$ .

*First term.* Write  $\sigma = \sigma_{\leq N} \sigma_{>N}$ ; the factor  $\sigma_{>N}$  is independent of  $\mathcal{F}_N$  and of  $\sigma_{\leq N}$  with mean  $\Pi_N = \prod_{\ell \in R, \ell > N} m_\ell$ , so  $2\langle E_N h, \sigma \rangle = 2\Pi_N \langle E_N h, \sigma_{\leq N} \rangle \leq 2\Pi_N \|h\|_\infty \leq w_0 \Pi_N$ .

*Second term.* Since  $\nu_{\text{sym}}$  is a product measure, tensorization of the variance gives  $\|h - E_N h\|_2^2 = \mathbb{E}[\text{Var}(h \mid \mathcal{F}_N)] \leq \sum_{\ell > N} \mathbb{E}[\text{Var}_{v_\ell}(h)]$ , and by Lemma 6.1,  $\text{Var}_{v_\ell}(h) \leq C(t_\ell) \mathbb{E}_{v_\ell} |\Delta_\ell h|^2$ . Now  $\Delta_\ell h$  coincides with  $\Delta_{\ell,k} h$  on  $\{v_\ell = k\}$  and  $\chi_3$  does not depend on  $v$ , so  $|\Delta_{\ell,k} h| = |\Delta_{\ell,k} f| \leq w_{\ell,k} L(f) \leq w_{\ell,k}$  there, whence  $\mathbb{E}_{v_\ell} |\Delta_\ell h|^2 \leq \sum_k (1 - t_\ell) t_\ell^k w_{\ell,k}^2 \leq \sum_k w_{\ell,k}^2$ . For  $\ell \geq 2$  and  $\beta \geq \beta_c$  we have  $t_\ell \leq 2^{-\beta_c} =: t_0 < 1$ , so  $C(t_\ell) \leq 4t_0/(1-t_0)^2 =: C_0$  uniformly, and

$$\|h - E_N h\|_2^2 \leq C_0 \sum_{\ell > N} \sum_k w_{\ell,k}^2 =: \rho_N^2.$$

Since  $\sum_{\ell,k} w_{\ell,k}^2 \leq (\sup_j w_j) \Sigma < \infty$ , we get  $\rho_N \rightarrow 0$  as  $N \rightarrow \infty$ , uniformly in  $\beta \geq \beta_c$ . As  $\|\sigma\|_2 = 1$ , Cauchy–Schwarz gives  $2\langle h - E_N h, \sigma \rangle \leq 2\rho_N$ .

Combining,  $W_1(\varphi_+, \varphi_-) \leq w_0 \Pi_N(\beta) + 2\rho_N$  for every  $N$ . Given  $\varepsilon > 0$ , choose  $N$  with  $2\rho_N < \varepsilon/2$ ; for that fixed  $N$ ,  $\sum_{\ell \in R, \ell > N} \ell^{-\beta} \uparrow \sum_{\ell \in R, \ell > N} \ell^{-\beta_c} = \infty$  as  $\beta \downarrow \beta_c$  by monotone convergence and

the divergence hypothesis, so  $\Pi_N(\beta) \rightarrow 0$  by Lemma 5.1 and  $w_0\Pi_N(\beta) < \varepsilon/2$  for  $\beta$  close enough to  $\beta_c$ .  $\square$

*Remark 6.3* (First order versus second order).

|                                                                        | $\beta \leq \beta_c$ | $\beta > \beta_c$                |
|------------------------------------------------------------------------|----------------------|----------------------------------|
| number of extreme $\text{KMS}_\beta$ states                            | 1                    | 2 (jump)                         |
| rank of $\Xi_\beta$ [9]                                                | 0                    | 1 (jump)                         |
| index $[M_{\text{sym}} : M_\varepsilon]$ (Paper II)                    | 1                    | 2 (jump)                         |
| $S(\omega_{\varphi_\pm} \  \omega_{\varphi_{\text{sym}}})$ (Paper III) | 0                    | $\log 2$ (jump)                  |
| $\text{diam}_{W_1}(\text{KMS}_\beta)$                                  | 0                    | $> 0$ (continuous at $\beta_c$ ) |

The first four count phases and jump by construction. The fifth does not, and Theorem 4.2 is why: were  $\Psi$  Lipschitz, the test function  $\Psi/L(\Psi)$  would give  $W_1 \geq 2/L(\Psi)$ , a bound not degenerating at the threshold, and the metric transition would also be of first order. Because  $\Psi$  is not Lipschitz the supremum is carried by functions measurable with respect to finitely many coordinates, whose correlation with  $\Psi$  is  $\Pi_N$ , and that correlation collapses as the Kakutani product degenerates. The failure of the order parameter to be Lipschitz is not a defect of the construction but the mechanism that makes the metric transition continuous.

*Remark 6.4* (A model computation). For  $R = \{r\}$  a single prime and constant weights  $w_0, w_r$  along the  $r$ -line — for which (2.3) fails and  $C(X)$  is not a compact quantum metric space, but the reduction of Lemma 5.2 still applies and  $L(f) = \sup_k (4w_0^{-2}h(k)^2 + w_r^{-2}|h(k+1) - h(k)|^2)^{1/2}$  for odd  $f = \chi_3 h$  with  $h$  a function of  $v_r$  — the distance is the convex program

$$W_1 = \sup \left\{ 2 \sum_{k \geq 0} (1-t)t^k (-1)^k h(k) : 4w_0^{-2}h(k)^2 + w_r^{-2}(h(k+1) - h(k))^2 \leq 1 \quad \forall k \right\},$$

which can be solved numerically after truncation. Two feasible profiles are explicit: the constant  $h \equiv w_0/2$ , which is the lower bound  $w_0 m_r$  of (5.2), and the alternating  $h(k) = (-1)^k a$  with  $a = \frac{1}{2}(w_0^{-2} + w_r^{-2})^{-1/2}$ , which gives  $2a = (w_0^{-2} + w_r^{-2})^{-1/2}$  for every  $t$ . With  $w_0 = 1$  (code in `code/verify_series.py`):

| $t$ | $w_r$ | $W_1$ (numerical) | alternating | $w_0 m_r$ (lower) | $w_0$ (upper) |
|-----|-------|-------------------|-------------|-------------------|---------------|
| 0.5 | 1     | 0.707             | 0.707       | 0.333             | 1             |
| 0.9 | 1     | 0.707             | 0.707       | 0.053             | 1             |
| 0.1 | 1     | 0.838             | 0.707       | 0.818             | 1             |
| 0.5 | 5     | 0.981             | 0.981       | 0.333             | 1             |
| 0.9 | 5     | 0.981             | 0.981       | 0.053             | 1             |

The alternating profile is the optimizer whenever its value exceeds the constant one; for small  $t$ , where  $m_r$  is close to 1, the optimizer interpolates between the two. The squeeze (5.2) holds in every case, and its lower bound is far from sharp for  $t$  near 1. This model has  $\sum_{\ell \in R} \ell^{-\beta} = t < \infty$  for all  $\beta$ , hence no transition, and cannot probe Theorem 6.2: the collapse there requires  $R$  infinite and is driven by  $\rho_N \rightarrow 0$ , which has no analogue for  $|R| = 1$ . We record the table only as a check on (5.2). In Lemma 6.1 the quantity  $B_n = t(1 - t^n)/(1 - t)^2$  increases to  $B = t/(1 - t)^2$ , and  $h(k) = k$  attains it.

## 7. ASSESSMENT

| Statement                                    | Status                     | Reference          |
|----------------------------------------------|----------------------------|--------------------|
| modular direction metrically blind           | true                       | Prop. 2.1          |
| horizontal distance on the base              | defined                    | Rem. 2.2           |
| $(C(X), L)$ a compact quantum metric space   | $\iff \sum_j w_j < \infty$ | Thm. 3.2, Rem. 3.3 |
| $\Psi$ Lipschitz                             | never, given the above     | Thm. 4.2           |
| $w_0 \Pi_\infty \leq W_1 \leq w_0$           | true                       | Thm. 5.3           |
| $W_1 > 0 \iff \sum_R \ell^{-\beta} < \infty$ | true                       | Thm. 5.3           |
| metric transition                            | continuous                 | Thm. 6.2           |

Three questions remain. The infinite-index prime inclusions are not the issue here, but the exact value of  $W_1(\varphi_+, \varphi_-)$  is unknown: Theorem 5.3 gives  $w_0 \Pi_\infty \leq W_1 \leq w_0$  and Remark 6.4 shows both bounds can be far from the truth; the upper bound  $w_0 \Pi_N + 2\rho_N$  from the proof of Theorem 6.2 suggests the answer is  $\inf_N (w_0 \Pi_N + 2\rho_N)$  up to constants, and it would be interesting to know whether the optimizer is asymptotically a finite-level function  $\Psi_N^{(m)}$ . Second, we have restricted to  $C(X)$ : the operator  $D_{\text{base}}$  is not defined on a representation of  $\mathcal{A}_{K,S}$ , and Remark 2.2 explains why the summable level weights that Theorem 3.2 requires are incompatible with bounded commutators with the isometries, while by Proposition 2.1 the modular operator alone gives infinite distances. Constructing a Dirac operator on  $\mathcal{A}_{K,S}$  with a nondegenerate vertical component — bounded commutators with the isometries, compact resolvent, Lip-kernel  $\mathbb{C}1$  — so that the distance on the  $\text{KMS}_\beta$  simplex is a genuine Connes distance of the crossed product, is open for Ore semigroup crossed products of this type; a twisted spectral triple in the sense of Connes–Moscovici, with twist the modular automorphism  $\sigma_{i\beta}$ , is the natural candidate, and  $\Psi$  might be twisted-Lipschitz although it is not Lipschitz. Third, for the cascade systems of [9] each generator  $\chi_n$  has its own elementary move and the analogue of Theorem 5.3 should give a family of bounds indexed by  $n$ ; whether  $\text{diam}_{W_1}$  then reproduces a continuous version of the weighted staircase, and whether it is continuous at every threshold, we do not know.

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